# PRANESH U S

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## Education

**Sri Krishna College of Engineering and Technology** 2024 – 2028
*Bachelor of Technology in Computer Science and Business Systems* CGPA: 8.21

## Experience

**Software Engineering Intern** Jan 2026 – Mar 2026
*Infosys Springboard*

- Fortified the MedVault healthcare platform by integrating Spring Security, JWT, and RBAC to secure REST APIs and MySQL databases
- Architected a refresh-token authentication flow with blacklist-based logout, reducing unauthorized access vulnerabilities by 90%
- Designed normalized MySQL schemas for patient record retrieval, cutting query latency by 30% for 10,000+ records

**Independent Software Developer** Jun 2025 – Present
*Self-Initiated Engineering Projects*

- Architected and deployed 3 production-grade distributed systems focused on AI governance, distributed state debugging, and zero-trust authentication
- Used Java 21, Spring Boot 4, Apache Kafka, Redis, and PostgreSQL to build scalable, enterprise-ready backend architectures

## Projects

**VPC-Native AI Gateway – Secure BPO Infrastructure** 2026
*Java 21, Spring Boot, React.js, Redis, Kafka, Microsoft Presidio*

- Engineered a VPC-native AI gateway integrating Microsoft Presidio for real-time PII masking and Shannon Entropy-based model routing
- Orchestrated a SHA-256 blockchain audit ledger and Kafka event streaming to ensure strict DPDP Act compliance for healthcare data
- Built a Redis-backed micro-billing ledger and dual-filter security scanner, reducing external AI API costs by 40% while maintaining sub-50ms latency

**Project Chronos – Distributed Time-Travel Debugger** 2026
*Java 21, Spring Boot, Apache Kafka, Redis, PostgreSQL, React.js*

- Developed a production-grade time-travel debugger for distributed backend systems, currently used in real-world scenarios to trace complex state mutations
- Engineered an event-sourcing architecture using Kafka and Redis to capture and replay distributed state, enabling deterministic backward and forward execution tracing
- Built an interactive React.js frontend to visualize state timelines and pinpoint root causes, reducing distributed system debugging time by 45%

**InstaAuth – Production-Grade Authentication Backend** 2025 – 2026
*Java 21, Spring Boot 4, Spring Security 7, PostgreSQL, Redis, Bucket4j*

- Architected a scalable authentication backend with JWT-based login, refresh-token rotation, Redis-backed token blacklisting, and OTP-based email verification
- Integrated enterprise-grade security features including per-IP rate limiting (Bucket4j), automated account lockout, and secure password reset flows
- Deployed a fully dockerized stack (PostgreSQL 16, Redis 7, MailHog) with Flyway schema migrations, ensuring zero-downtime deployments and environment parity

## Technical Skills

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|---|---|
| **Languages:** | Java, JavaScript, Python, SQL, HTML, CSS |
| **Frameworks:** | Spring Boot 4, Spring Security 7, React.js, Tailwind CSS, FastAPI |
| **Backend/Infra:** | REST APIs, JWT, OAuth2, Redis, Apache Kafka, Docker, VPC Architecture, Microservices |
| **Database/Cloud:** | PostgreSQL, MySQL, Firebase, AWS (EC2, IAM, S3), Git, GitHub, Postman |
| **Concepts:** | Authentication & Authorization, RBAC, API Security, Event Sourcing, Rate Limiting, Database Design, Backend Development |
| **Core:** | Data Structures & Algorithms, OOP, Distributed Systems, Zero-Trust Security, DBMS, Computer Networks |

## Certifications

Claude Code in Action (2026) · Introduction to LLMs – Google Cloud (May 2025) · Python for Data Science – IBM (Aug 2025) · Software Engineering, Intro to AI, DB & SQL – Infosys (2026) · Networking Basics – Cisco (Jan 2025)